SafeSea: SDG-Based Smart Maritime Boundary Protection and Environmental Monitoring System Using Object-Oriented Analysis and Design

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

CSA1110- OBJECT ORIENTED ANALYSIS AND DESIGN

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "SafeSea: SDG-Based Maritime Boundary Protection and Environmental Monitoring System Using Object Oriented Analysis And Design" has been carried out by Tejaswini. M (192411079) under the supervision of Dr Manimaran A and is submitted in partial fulfilment of the requirements for the current semester of the B.E Computer Science and Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, Tejaswini. M of the Department of CSE, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled "SafeSea: SDG-Based Maritime Boundary Protection and Environmental Monitoring System Using Object Oriented Analysis And Design" is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity.All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Total	—	—	—	—	100%

ABSTRACT

SafeSea is a smart maritime safety and navigation system designed to support fishermen in safer route planning, boundary monitoring, and emergency response. Fishermen may face difficulties in identifying restricted maritime areas, monitoring vessel positions, selecting safer routes, and responding quickly to hazardous situations. These challenges can increase navigation risks and may lead to accidental entry into restricted zones.

The proposed system addresses these problems through an object-oriented software design that integrates fisherman registration, voyage management, navigation monitoring, safe route management, maritime boundary and hazard detection, emergency communication, and navigation analytics. The system follows Object-Oriented Analysis and Design (OOAD) principles to identify system actors, objects, classes, relationships, and interactions. Use case, class, activity, and other design models are used to represent the system structure and behaviour before implementation.

The system provides a centralized platform where fishermen can register their details, manage voyages, monitor their navigation position, and receive boundary or hazard alerts. It also supports emergency communication by transmitting vessel location and emergency information to appropriate response personnel. Navigation analytics and reporting features provide useful information for reviewing voyage activities and system performance.

The developed system was tested through functional and workflow-based verification of major features, including registration, authentication, navigation monitoring, boundary alerts, emergency communication, and analytics. The testing confirmed that the major system functions operated according to their intended requirements.

Overall, SafeSea demonstrates how OOAD can be applied to develop a structured maritime navigation and safety solution. The system provides an integrated approach to navigation assistance, boundary awareness, hazard notification, and emergency support, while establishing a foundation for future enhancements such as real-time data integration, improved route optimization, GIS-based visualization, and mobile support.

Keywords : SafeSea, Fisherman Navigation, Maritime Safety, Boundary Monitoring, OOAD, Emergency Communication, Hazard Detection

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LIST OF ABBREVIATIONS / SYMBOLS

Abbreviation / Symbol	Description	Unit
OOAD	Object-Oriented Analysis and Design	—
GPS	Global Positioning System	—
GNSS	Global Navigation Satellite System	—
SOS	Save Our Souls / Emergency Signal	—
GIS	Geographic Information System	—
UI	User Interface	—
API	Application Programming Interface	—
RBAC	Role-Based Access Control	—
d	Distance between two geographical points	km
R	Earth's radius	km
ϕ	Latitude	Degrees
λ	Longitude	Degrees

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Maritime fishing involves continuous movement of vessels through open water, where fishermen need to know their location, follow safe routes, and avoid restricted or hazardous areas. Manual monitoring can make it difficult to identify boundary violations and respond quickly during emergencies.

SafeSea is a smart maritime navigation and safety system designed to support fishermen during voyages. The system combines fisherman registration, voyage management, navigation monitoring, safe route management, boundary and hazard detection, emergency communication, and navigation analytics. The system is designed using Object-Oriented Analysis and Design (OOAD) principles to represent users, system objects, interactions, and relationships clearly.

1.2 Need for the Project

Fishermen may face navigation risks such as entering restricted maritime areas, encountering hazardous conditions, or experiencing emergencies during voyages. A system is therefore needed to provide timely navigation information and safety support.

The main limitations of existing manual approaches are:

- Difficulty in continuously monitoring vessel location.
- Risk of entering restricted or unsafe areas.
- Limited support for safe route planning.
- Delayed emergency communication.
- Difficulty in maintaining voyage history and analysing navigation activities.

SafeSea provides an integrated software solution to improve navigation awareness, boundary monitoring, and emergency response.

Existing Problem	**SafeSea Solution**
Difficulty monitoring vessel location	Navigation monitoring
Entry into restricted areas	Boundary detection and alerts
Unsafe route selection	Safe route management
Delayed emergency communication	SOS and emergency alerts
Limited voyage records	Voyage management and history

Difficulty reviewing navigation activities	Analytics and reporting

Table 1.1: Existing Problems and Proposed SafeSea Solution

1.3 Problem Statement

Fishermen require a reliable system to manage voyages, monitor vessel locations, identify restricted or hazardous areas, select safer routes, and communicate during emergencies. The engineering challenge is to integrate these functions into a single system while maintaining usability, accurate location-based monitoring, timely alerts, secure access, and reliable system operation under practical navigation conditions.

1.4 Project Aim

The aim of SafeSea is to design and develop an object-oriented maritime navigation and safety system that supports fishermen through navigation monitoring, safe route management, boundary and hazard detection, emergency communication, and navigation analytics.

1.5 Project Objectives

1. To design a structured maritime navigation system using OOAD principles.
2. To develop fisherman registration and voyage management functions.
3. To implement navigation monitoring and safe route management.
4. To provide boundary and hazard detection with appropriate alerts.
5. To implement emergency communication and SOS support.
6. To develop navigation analytics and reporting features.
7. To test and evaluate the major system functions.

Objective	**Expected Outcome**
Design using OOAD	Structured system models
Develop registration and voyage management	User and voyage management
Implement navigation monitoring	Vessel location awareness
Detect boundaries and hazards	Safety alerts
Implement emergency communication	SOS and emergency support
Develop analytics	Navigation reports
Test the system	Verified system functions

Table 1.2: Objectives and Expected Outcomes

1.6 Scope

Included

- Fisherman registration and authentication.
- Voyage creation and management.
- Navigation and location monitoring.
- Safe route management.
- Maritime boundary and hazard detection.
- Warning and danger alerts.
- Emergency communication and SOS.
- Navigation analytics and reporting.
- OOAD-based system modelling and documentation.

Excluded

- Physical navigation equipment and hardware.
- Replacement of official maritime navigation systems.
- Complete autonomous vessel control.
- Large-scale commercial maritime deployment.

Assumptions

- Users provide valid registration and vessel information.
- Location and navigation data are available to the system.
- Users have access to the application during a voyage.

Boundary Conditions

The system is developed as a software-based decision-support solution. Its effectiveness depends on the availability and quality of navigation/location information and should not be considered a replacement for certified maritime safety systems.

1.7 Expected Engineering Outcome

The expected outcome is a functional SafeSea software prototype that integrates:

- Fisherman and voyage management.
- Navigation monitoring and safe route support.
- Boundary and hazard detection.
- Emergency communication and SOS.
- Navigation analytics and reporting.
- OOAD models representing system structure and behaviour.

The project demonstrates the application of Object-Oriented Analysis and Design to develop an integrated maritime navigation and safety solution.

CHAPTER 2
LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research papers, maritime navigation technologies, existing vessel monitoring systems, and software design approaches were reviewed to understand current solutions for navigation safety and maritime monitoring. The review mainly focused on vessel navigation, route management, boundary monitoring, hazard detection, emergency communication, and object-oriented system design. Recent research from 2025 and 2026 was given priority.

2.2 Review of Existing Work

1. Vessel Trajectory Data Mining

Troupiotis-Kapeliaris, Kastrisios and Zissis (2025) presented a review of vessel trajectory data mining techniques. The study highlights the importance of vessel movement data for understanding routes, detecting unusual behaviour, and supporting maritime decision-making. However, the work mainly focuses on trajectory data analysis rather than providing an integrated fisherman-oriented navigation and emergency-support system.

2. Vessel Trajectory Classification

Kutin et al. (2026) investigated vessel trajectory classification using a graph-based approach. The work demonstrates how vessel movement patterns can be classified for maritime applications. Its focus is primarily on trajectory classification, whereas SafeSea combines navigation support, boundary monitoring, emergency communication, and user management.

3. Multi-Source Maritime Navigation Data

Wang, Jiao and Pan (2026) proposed a vessel trajectory recognition approach based on multi-source data fusion. The study shows the value of combining different maritime information sources for better vessel trajectory recognition. However, the approach is more focused on trajectory recognition than on developing a complete user-oriented navigation safety application.

Existing Technologies

Modern maritime systems commonly use GPS, digital maps, geofencing, navigation databases, and communication technologies to monitor vessel positions and provide safety information. These technologies provide useful foundations for SafeSea, but integrating them into a simple application specifically supporting fishermen remains an important engineering requirement.

Object-Oriented Design

Object-Oriented Analysis and Design provides a structured method for identifying actors, objects, classes, relationships, and system interactions. SafeSea applies OOAD concepts to organize its navigation, voyage management, boundary monitoring, emergency communication, and analytics functions into a structured software design.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Manual navigation	Simple and inexpensive	Limited monitoring and alerts	Shows need for automated support
GPS-based navigation	Provides vessel location	Does not necessarily provide complete safety management	Used as a foundation for navigation monitoring
Vessel trajectory analysis	Supports route and movement analysis	Mainly focuses on trajectory data	Supports navigation analysis
Digital maritime monitoring	Enables location and boundary monitoring	May require specialized systems	Relevant to boundary detection
Emergency communication systems	Supports emergency response	Often separate from navigation functions	Integrated into SafeSea
Safe route applications	Helps users plan routes	May not include emergency and boundary functions	Provides route-management concept
SafeSea	Integrates navigation, boundary alerts, emergency support and analytics	Prototype-level system with scope limitations	**Proposed integrated solution**

2.4 Research / Engineering Gap

The reviewed approaches provide useful solutions for individual maritime functions such as vessel tracking, trajectory analysis, route planning, and emergency communication. However, these functions are often considered separately.

The identified engineering gap is the lack of a simple integrated system designed around fishermen's navigation needs that combines navigation monitoring, safe route management, maritime boundary and hazard detection, emergency communication, and navigation analytics in one application.

2.5 Proposed Contribution

SafeSea addresses the identified gap by providing an integrated software-based maritime navigation and safety system. The system applies Object-Oriented Analysis and Design to structure its users, functions, classes, and interactions.

The main contribution is the integration of:

- Fisherman registration and voyage management.
- Navigation and safe route management.
- Maritime boundary and hazard detection.
- Emergency communication and SOS support.
- Navigation analytics and reporting.
- OOAD models for structured system development.

This approach provides a single, organized platform for supporting fishermen's navigation and safety

CHAPTER 3
ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Fisherman	Register personal, fishing license, and vessel details; manage voyages; monitor vessel location; view safe routes; receive boundary and hazard alerts; and send SOS alerts during emergencies.
Coast Guard	Monitor vessel locations, receive boundary and hazard alerts, receive SOS/emergency information, and access relevant vessel and voyage details for response activities.
Administrator	Manage user and vessel information, control user access, maintain system records, and view navigation and system reports.

Table 3.1: Stakeholder / User Requirements

Functional & Non-Functional Requirements

Requirement	Type	Measurable Target
User registration and login	Functional	Registration/login completed successfully
Voyage creation and management	Functional	Voyage details stored and retrievable
Vessel location monitoring	Functional	Current location displayed when data is available
Boundary detection	Functional	Alert generated when restricted boundary is approached

Hazard detection	Functional	Hazard warning displayed for detected risk
Safe route management	Functional	Suitable route displayed based on available navigation data
SOS communication	Functional	Emergency message contains vessel location and status
Navigation history	Functional	Previous voyage records available for review
System response time	Non-Functional	Major user actions respond within 3 seconds under normal conditions
Security	Non-Functional	Authentication and role-based access implemented
Usability	Non-Functional	Main functions accessible through a simple user interface
Reliability	Non-Functional	System maintains consistent records during normal operation

Table 3.2: Functional and Non-Functional Requirements

3.2 Constraints & Applicable Standards

Applicable Constraints

The major constraints applicable to SafeSea are:

- **Technical:** System performance depends on the availability and accuracy of GPS/navigation data and network connectivity.
- **Economic:** The project is developed as a software prototype using commonly available development tools and infrastructure.
- **Environmental:** The system supports safer navigation and reduces unnecessary route deviations and travel.
- **Social:** Alerts and navigation information must be presented clearly so that fishermen can understand them quickly.
- **Health & Safety:** The system provides decision support for navigation and emergencies but does not replace official maritime safety procedures.
- **Ethical:** User and vessel information must be protected and used only for authorized

purposes.

- **Security:** Authentication, authorization and controlled access are required to protect system information.

Applicable Standards

Standard / Code	Requirement	How Applied in This Project
ISO/IEC 25010	Defines software quality characteristics such as usability, reliability, security and performance.	Used as a reference while designing and evaluating SafeSea.
ISO/IEC 27001	Provides requirements for information security management.	Security principles are considered for protecting user and vessel information.
IEEE 1012	Provides guidance for software verification and validation.	Functional testing and verification are performed for major system features.
SOLAS	Provides international maritime safety requirements.	Used as a maritime safety reference; SafeSea does not claim regulatory compliance.

Table 3.3: Constraints and Applicable Standards

3.3 Design Specifications

Parameter	Target/Requirement	Unit	Priority
User authentication	Secure login and access control	—	High
Vessel location	Display available vessel position	Coordinates	High
Boundary monitoring	Detect restricted-area proximity	Distance/zone	High
Alert generation	Generate warning for detected risks	Seconds	High

Route management	Provide suitable navigation route	Route	High
SOS communication	Send emergency information with location	Message	High
Response time	Major operations within 3 seconds under normal conditions	Seconds	Medium
Data availability	Maintain accessible voyage records	Records	Medium
Usability	Simple and understandable interface	—	High
Security	Role-based access and authentication	—	High

Table 3.4: Design Specifications

3.4 Alternative Solutions & Evaluation

Alternative 1: Basic GPS-Based Navigation System

A basic GPS navigation system provides vessel positioning and route information. It mainly focuses on location tracking and navigation without integrating user management, boundary alerts and emergency support.

Alternative 2: Separate Maritime Monitoring Applications

Multiple specialized applications can be used for registration, navigation, boundary monitoring and emergency communication. This provides individual functionality but requires users to operate and maintain several systems.

Alternative 3: SafeSea Integrated OOAD-Based System

SafeSea combines user management, voyage management, navigation monitoring, route management, boundary and hazard alerts, emergency communication and analytics in one structured software system. OOAD models are used to organize the system objects, classes, relationships and interactions.

Evaluation Matrix

Scoring: 1 = Poor, 5 = Excellent

Criterion	Weight	Alternative	Alternative	Alternative

		1	2	3
Performance	25%	4	4	5
Cost	15%	5	3	4
Safety	25%	2	4	5
Sustainability	10%	4	3	5
Reliability	25%	3	4	5
Weighted Score	100%	3.45	3.70	4.85

Table 3.5: Alternative Solutions Evaluation Matrix

Selected solution: Alternative 3 – SafeSea Integrated OOAD-Based System

3.5 Selected Design & Detailed Design

Final Design Selection

Alternative 3 was selected because it provides the highest overall evaluation score. Unlike a basic GPS system or multiple separate applications, SafeSea integrates navigation, boundary monitoring, emergency communication and user management into a single software platform. The OOAD approach provides a structured representation of system actors, objects, classes and interactions, making the design easier to maintain and extend.

SafeSea System Architecture

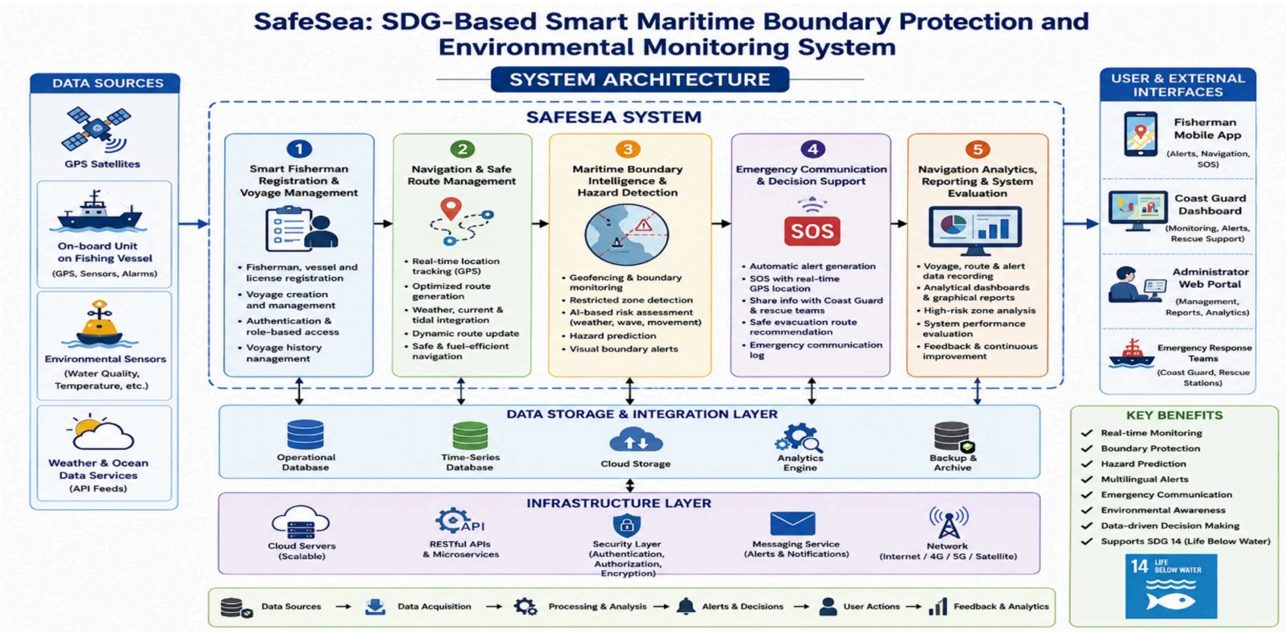


Figure 3.1: SafeSea System Architecture

The system can be organized into the following layers:

User Interface → Application/Business Logic → Navigation & Safety Services → Database

- The User Interface allows fishermen, Coast Guard personnel and administrators to interact with the system.
- The Application Layer manages authentication, user requests, voyage information and system operations.
- Navigation and Safety Services process location information, route information, boundary conditions and emergency events.
- The Database stores user, vessel, voyage, navigation and emergency records.

Systems Approach

The system components operate together through defined software interactions. User and vessel information is maintained before voyage operations begin. Navigation services provide location and route information, while boundary and hazard functions evaluate the navigation status and generate alerts when required. Emergency communication uses the available vessel and location information to support SOS notifications and response activities.

Essential Engineering Calculation

For boundary monitoring, the distance between the vessel and a defined boundary point can be estimated using geographical coordinates. A suitable distance calculation can be used to determine whether the vessel is approaching a restricted zone.

For example, using the Haversine equation:

$$d = 2Rsin^{-1}\left(\sqrt{sin^{2}\frac{\Delta\phi}{2} + \cos cos\,\phi_{1}\ \cos cos\,\phi_{2}\ sin^{2}\frac{\Delta\lambda}{2}}\right)$$

where:

- d= distance between two geographical points
- R= Earth's radius
- ϕ= latitude
- λ= longitude

The calculated distance can be compared with a predefined safety threshold to generate a warning.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Integrated application	Separate applications	Centralized access and simpler workflow	Higher initial design effort	Selected because major safety functions are available in one system
OOAD architecture	Procedural design	Clear objects, classes and relationships	Requires detailed modelling	Selected because the project follows OOAD methodology
GPS-based monitoring	Manual position entry	More accurate and automated monitoring	Depends on location data availability	Selected because automated monitoring improves safety awareness
Role-based access	Common access for all users	Better security and controlled operations	Requires access-control design	Selected to separate Fisherman, Coast Guard and Administrator access
Email/SOS alerts	Manual emergency reporting	Faster communication	Depends on network/service availability	Selected for quicker emergency notification
Centralized voyage records	Manual records	Easy retrieval and analysis	Requires database storage	Selected for reliable voyage history

Table 3.6: Trade-off Analysis

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence	Impact on

		Evaluated	Final Decision
Sustainability	Reduce unnecessary navigation and improve route planning	Route-management and navigation requirements	Integrated route support was selected
Ethics	Protect fisherman and vessel information	User authentication and access requirements	Role-based access and controlled data handling were included
Health & Safety	Provide timely awareness of boundaries and hazards	Boundary and emergency requirements	Warning and SOS features were included
Security	Prevent unauthorized access to system information	Authentication and role requirements	Secure login and role-based authorization were included

Table 3.7: Sustainability, Ethics, Safety and Security Considerations

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
GPS/location data unavailable	Medium	High	High	Display data status and avoid relying on unavailable location information	Medium
Network connectivity failure	Medium	High	High	Provide clear connection status and retain available records	Medium
Incorrect boundary information	Low	High	High	Use validated boundary data and regular data verification	Low

False or delayed alert	Medium	High	High	Test alert conditions and maintain clear warning thresholds	Medium
Unauthorized user access	Medium	High	High	Authentication and role-based access control	Low
Database failure	Low	High	Medium	Regular backup and error handling	Low
User misunderstanding of warning	Medium	High	High	Use clear messages, warning indicators and simple interface design	Medium
System software failure	Low	Medium	Medium	Functional testing, validation and error handling	Low

Table 3.8: SafeSea Risk Register

Security Measures

SafeSea incorporates basic security measures to protect system information:

1. **Authentication –** users must log in before accessing protected functions.
2. **Role-Based Access Control –** access is provided according to the user's role.
3. **Input Validation –** user inputs are checked before processing.
4. **Protected User Data** – personal and vessel information is not exposed to unauthorized users.
5. **Session Management –** authenticated access is controlled through appropriate session mechanisms.
6. **Error Handling –** system errors are handled without exposing unnecessary internal information.

Safety Consideration

SafeSea is designed as a decision-support and navigation-awareness system. Its alerts and route recommendations assist users but should not be treated as a replacement for official maritime navigation equipment, Coast Guard instructions, or established emergency procedures. This limitation is important because navigation decisions may depend on real-time conditions and external information availability.

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

SafeSea was developed using an object-oriented software development approach based on Object-Oriented Analysis and Design (OOAD). The system was first analysed by identifying actors, requirements, use cases, classes, objects, and their relationships. The design was then converted into software components for user management, voyage management, navigation, safety monitoring, emergency communication, and reporting. The developed functions were tested individually and as complete workflows to verify that they satisfy the defined requirements.

Resources Used

Resource	Purpose
OOAD models	System analysis and design
GPS/Location data	Vessel position monitoring
Digital map/navigation data	Route and navigation support
Database	Storage of user, vessel, voyage and alert records
Web interface	User interaction and system monitoring
Development environment	Software implementation and testing

Table 4.1: Resource Used

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Software Implementation

SafeSea is primarily a software-based system. The implementation focuses on providing a user interface, application logic, database management, navigation monitoring, boundary detection, alerts, emergency communication, and analytics.

No physical fabrication or dedicated hardware prototype is required for the software implementation. Navigation information is handled through available location and map data.

Tools Used

Tool / Software / Equipment	Purpose	Stage Used

VS Code	Source-code development and editing	Implementation
React.js	Development of the user interface	Implementation
JavaScript	Application functionality and interaction	Implementation
Flask	Backend/API development	Implementation
MongoDB	Storage of system records	Implementation
Git/GitHub	Version control and source-code management	Development
Browser Developer Tools	Interface debugging and testing	Testing
Postman	API testing and verification	Testing

Table 4.2: Tools Used

Implementation Evidence

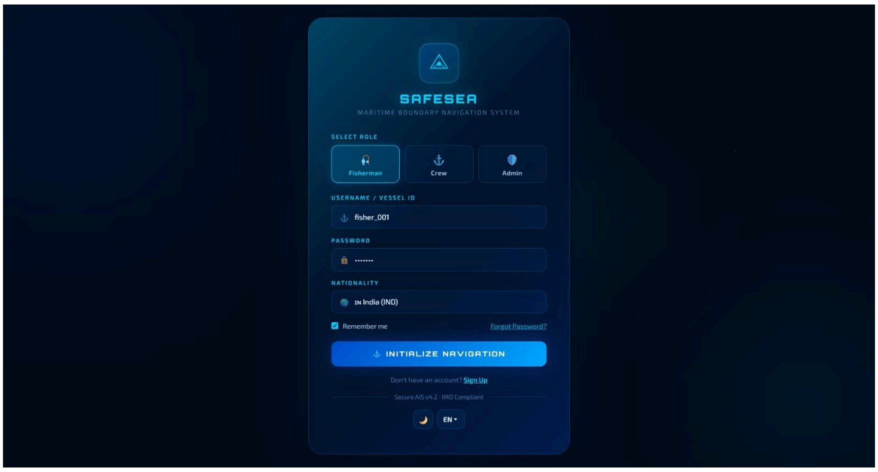


Figure 4.1: Login Page

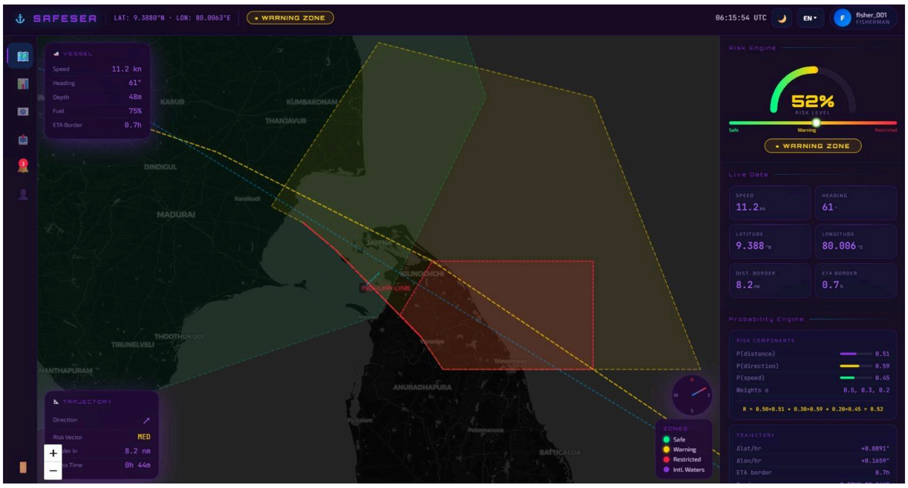


Figure 4.2: Dashboard Page

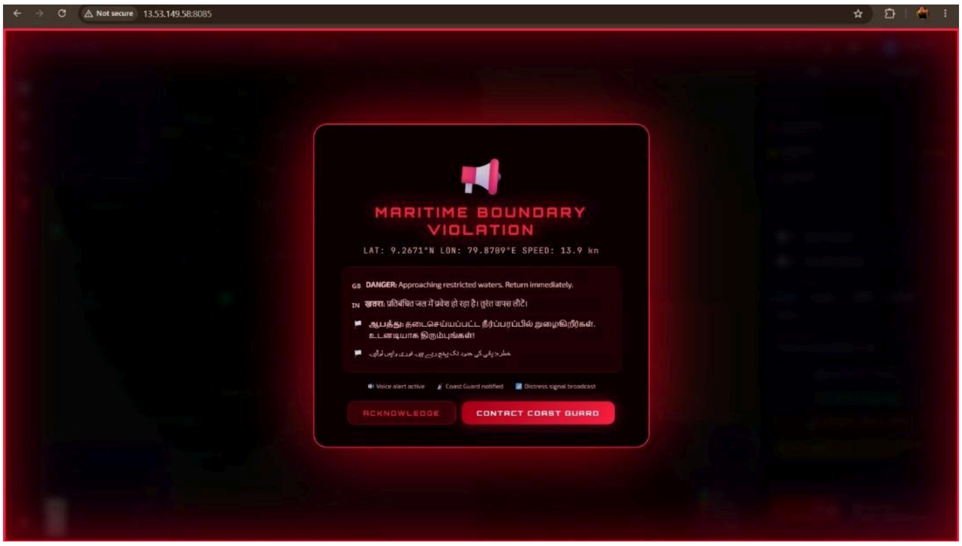


Figure 4.3: Danger Alert

4.3 Test Plan & Verification

Test Plan

The SafeSea system was tested using functional and workflow-based testing. Each major requirement was mapped to a test method and an acceptance criterion. Testing was performed to verify registration, authentication, voyage management, navigation monitoring, boundary detection, emergency communication, and reporting functions.

Functional Test Plan

Requirement	Test Method	Acceptance Criterion
User registration	Enter valid user details	User account created

	and submit	successfully
User login	Enter valid credentials	User is authenticated successfully
Voyage management	Create and update voyage details	Voyage information stored correctly
Location monitoring	Provide valid location data	Vessel position displayed correctly
Boundary detection	Test vessel position near restricted area	Appropriate warning generated
Hazard detection	Provide hazard condition	Hazard warning displayed
Safe route management	Select navigation destination	Suitable route information displayed
SOS communication	Trigger emergency function	SOS information generated successfully
Navigation history	Open previous voyage records	Stored voyage information displayed
Role-based access	Login using different user roles	Only authorized functions are accessible

Table 4.3: Functional Test Plan

Specification Verification

Specification	**Required Value**	**Achieved Value**	**Status**
User authentication	Successful authentication	Successfully implemented	Pass
User registration	Valid details stored	Successfully implemented	Pass
Voyage management	Create/update voyage	Successfully implemented	Pass
Navigation monitoring	Vessel position displayed	Successfully implemented	Pass
Boundary detection	Warning for restricted area	Successfully implemented	Pass

Hazard notification	Hazard warning displayed	Successfully implemented	Pass
SOS communication	Emergency information generated	Successfully implemented	Pass
Role-based access	Authorized access only	Successfully implemented	Pass
Navigation records	Previous records retrievable	Successfully implemented	Pass

Table 4.4: Specification Verifiction

4.4 Results

The developed SafeSea prototype was evaluated using functional testing of the major system operations. The tests confirmed that the implemented functions operated according to their intended requirements.

Test Results Summary

Test Category	**Tests Performed**	**Passe d**	**Status**
Registration & Login	2	2	Pass
Voyage Management	2	2	Pass
Navigation Monitoring	2	2	Pass
Boundary & Hazard Detection	2	2	Pass
Emergency Communication	1	1	Pass
Access Control	1	1	Pass
Total	10	10	Pass

Table 4.5: Test Result Summary

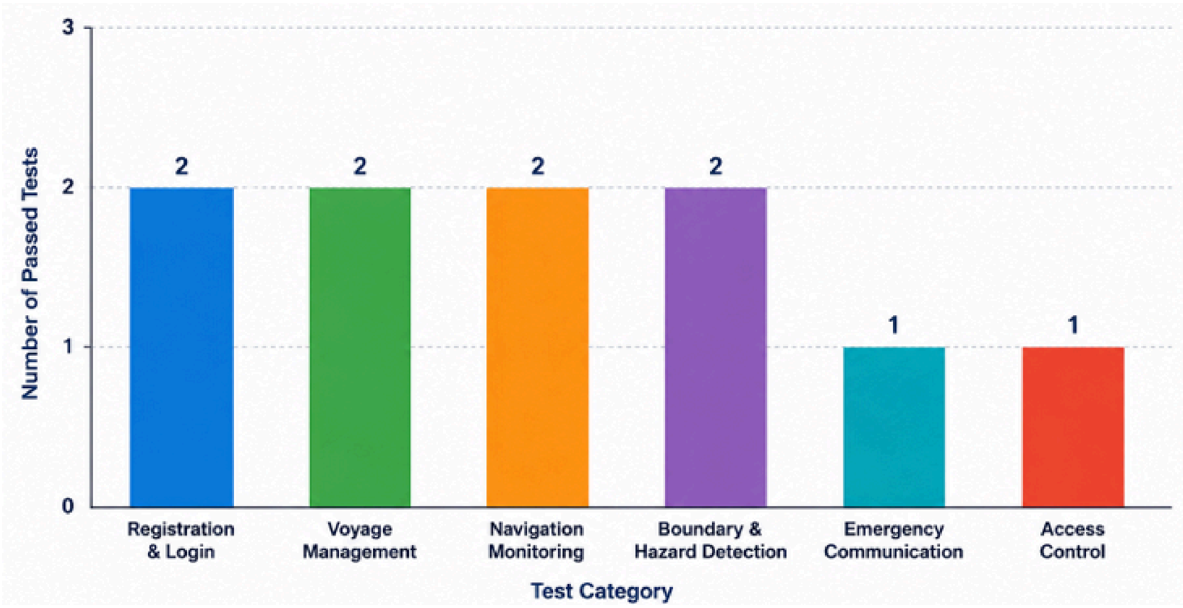


Figure 4.4: SafeSea Functional Testing Results

4.5 Data Analysis & Engineering Judgment

The testing results indicate that the major SafeSea functions operated according to their defined requirements. Registration and authentication successfully controlled user access, while voyage and navigation functions supported the management and monitoring of vessel activities. Boundary and hazard detection provided safety notifications, and the emergency function generated the required SOS information.

The results also indicate that SafeSea provides better integration than using independent functions separately. However, actual navigation accuracy depends on the quality and availability of external location, map, network, and maritime information. Therefore, the prototype should be considered a software decision-support system rather than a replacement for certified maritime navigation and emergency systems.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design the system using OOAD principles	Use case, class and activity models	Fully Achieved
Develop registration and voyage management	Registration, login and voyage functions	Fully Achieved
Implement navigation	Navigation/location interface	Fully

monitoring		Achieved
Provide boundary and hazard detection	Boundary and hazard alerts	Fully Achieved
Implement emergency communication	SOS/emergency functionality	Fully Achieved
Develop navigation analytics and reporting	Analytics and reporting interface	Fully Achieved
Test and evaluate system functions	Functional and workflow testing	Fully Achieved

Table 4.6: Achievement of Project Objectives

4.7 Strengths & Limitations

Strengths

- Integrates major maritime safety functions into one system.
- Provides vessel navigation and location awareness.
- Supports restricted-area and hazard alerts.
- Provides emergency/SOS communication support.
- Uses role-based access for different users.
- Maintains voyage information for later review.
- OOAD-based structure supports future system expansion.
- Provides a simple user-oriented interface.

Limitations

- System performance depends on GPS/location data availability.
- Network connectivity may affect real-time communication.
- The prototype does not replace certified maritime navigation systems.
- Accuracy depends on the quality of available map and boundary data.
- The current system has limited physical hardware integration.
- Large-scale real-world deployment requires additional validation.

4.8 Project Planning, Roles & Budget

Project Milestone Timeline

Phase	Major Activities
Phase 1	Problem identification and requirement analysis
Phase 2	Literature review and existing-system analysis
Phase 3	OOAD modelling and system design
Phase 4	Interface and backend development
Phase 5	Navigation, boundary and emergency functionality
Phase 6	Integration and database implementation
Phase 7	Testing and debugging
Phase 8	Final evaluation and report preparation

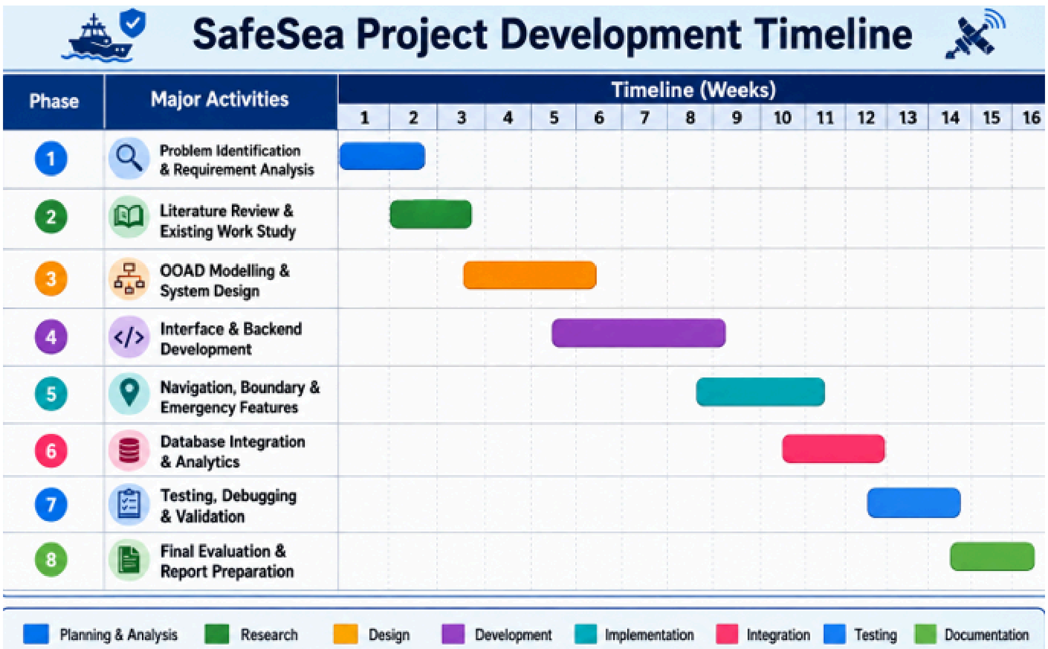


Figure 4.5: SafeSea Project Development Timeline

Team Roles and Contributions

Student	Assigned Responsibility	Actual Contribution
Tejaswini M (192411079)	Navigation and safe route management	Developed navigation interface, route-management functions and contributed to testing and documentation
Sharu Priya M (192411116)	Fisherman registration and voyage management	Developed registration, authentication and voyage-related functions and contributed to documentation
Vishalsai BJ (192320012)	Boundary monitoring and emergency communication	Developed boundary detection, alerts and emergency communication functions and contributed to testing
D Chaitanya (192320028)	Navigation analytics and system evaluation	Developed analytics/reporting functions and contributed to system evaluation, OOAD diagrams and documentation

Table 4.7: Team Roles & Contribution

Project Budget

Item	Estimated Cost	Actual Cost
Software Development Tools	₹500	₹0
Cloud/Database Services	₹500	₹0
Internet/Data	₹500	₹300
Prototype & Documentation	₹500	₹300
Total	₹2,000	₹600

Table 4.8: Project Budget

CHAPTER 5
CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

SafeSea was developed to address the challenges faced by fishermen in navigation, maritime boundary awareness, hazard identification, and emergency communication. The system provides an integrated software solution for fisherman registration, voyage management, navigation monitoring, safe route management, boundary and hazard detection, emergency communication, and navigation analytics.

The system was designed using Object-Oriented Analysis and Design (OOAD) principles. Use case, class, activity, and system design models were used to represent the users, system objects, relationships, and interactions. The implementation provided a structured platform where fishermen can manage voyage information, monitor navigation status, receive safety alerts, and initiate emergency communication when required.

Functional and workflow-based testing was performed for the major system operations. The testing results showed that the implemented functions operated according to their intended requirements, including registration, authentication, voyage management, navigation monitoring, boundary detection, alerts, emergency communication, and access control.

Overall, SafeSea demonstrates the application of OOAD principles to a practical maritime safety problem. The project provides a foundation for developing a more comprehensive maritime decision-support system with improved navigation awareness, safety monitoring, emergency response, and future real-time integration.

5.2 Future Work

The following improvements can be considered for future development:

- Integration with real-time GPS/GNSS devices for continuous vessel tracking.
- Integration of real-time weather, wave, tide, and ocean-current data.
- Improved route optimization using live maritime conditions.
- Integration with official maritime boundary and restricted-zone databases.
- Development of a dedicated mobile application for fishermen.
- Integration of additional emergency communication channels such as SMS and push notifications.
- Addition of offline functionality for areas with poor network connectivity.
- Integration with physical maritime safety equipment and vessel monitoring devices.

- Improved analytics for voyage history, route performance, and safety events.
- Deployment and evaluation with real-world maritime users under controlled conditions.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- Learned the application of Object-Oriented Analysis and Design (OOAD) in a real-world software project.
- Learned how to identify actors, use cases, classes, objects, relationships, and system interactions.
- Gained knowledge of navigation monitoring, boundary detection, and emergency communication workflows.
- Learned how to connect frontend, backend, and database components.
- Gained practical experience in software testing, debugging, documentation, and project management.

Engineering & Professional Skills Developed

- Requirements analysis and system modelling.
- Object-oriented software design.
- Full-stack software development.
- Database design and management.
- Functional testing and debugging.
- Technical documentation and report preparation.
- Team collaboration and task coordination.
- Problem-solving and engineering decision-making.
- Project planning and progress tracking.

Individual Reflection – Tejaswini M

Register No.: 192411079

The main challenge I encountered was designing the navigation and route-management functions in a way that was simple for users to understand. I addressed this by organizing the navigation workflow clearly and connecting it with the required location information. A key engineering decision was to provide navigation information through a structured interface rather than presenting complex information directly to the user. I independently improved my understanding of navigation interfaces and OOAD modelling. If the project were repeated, I would improve the real-time navigation and route optimization features.

Individual Reflection – Sharu Priya M

Register No.: 192411116

The most difficult problem I encountered was managing user registration, authentication, and voyage information consistently. This was addressed by separating user-related operations and applying controlled access to system functions. A key decision was to use role-based access so that different users could access only the functions relevant to them. I gained additional knowledge of authentication, database operations, and software requirements analysis independently. If the project were repeated, I would improve the authentication mechanism and provide more detailed user-management features.

Individual Reflection – Vishalsai BJ

Register No.: 192320012

The main engineering challenge was implementing boundary monitoring, safety alerts, and emergency communication in a reliable workflow. I addressed this by defining clear conditions for generating warnings and organizing emergency information so that it could be communicated effectively. A key decision was to include immediate alert generation when a safety condition was detected. I developed new knowledge about geofencing concepts, alert handling, and emergency communication workflows. If the project were repeated, I would improve real-time boundary monitoring and integrate additional communication methods.

Individual Reflection – D Chaitanya

Register No.: 192320028

The main challenge was organizing navigation analytics and evaluating the overall system in a structured manner. I addressed this by defining relevant records and presenting the required information through reports and analytical views. A key engineering decision was to use structured system data for reviewing navigation activities and evaluating system performance. I gained additional knowledge of software evaluation, analytics, OOAD diagrams, and technical documentation. If the project were repeated, I would improve the analytics with more detailed performance indicators and real-time reporting.

REFERENCES

A. Literature References

[1] A. Troupiotis-Kapeliaris, C. Kastrisios, and D. Zissis, "Vessel Trajectory Data Mining: A Review," *IEEE Access*, vol. 13, pp. 4827–4856, 2025, doi: 10.1109/ACCESS.2025.3525952.

[2] N. Kutin, R. Bucknall, P. Wu, and Y. Liu, "Vessel Trajectory Classification from a Graph Network Perspective," *Advanced Engineering Informatics*, vol. 69, Art. no. 104082, 2026, doi: 10.1016/j.aei.2025.104082.

[3] J. Wang, L. Jiao, and Q. Pan, "Vessel Trajectory Recognition Based on Multi-Source Data Multi-Stage Fusion," *IEEE Sensors Journal*, 2026, doi: 10.1109/JSEN.2026.3658076.

B. Standards

[4] ISO/IEC, *ISO/IEC 25010:2023 — Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — Product Quality Model*, ISO, Geneva, Switzerland, 2023.

[5] ISO/IEC, *ISO/IEC 27001:2022 — Information Security, Cybersecurity and Privacy Protection — Information Security Management Systems — Requirements*, ISO, Geneva, Switzerland, 2022.

[6] IEEE, *IEEE Standard for System, Software and Hardware Verification and Validation*, IEEE Std. 1012-2016, 2016.

C. Main Website/Software References

[7] Mozilla Developer Network, "JavaScript Guide," *MDN Web Docs*, 2026.

[8] React Team, "React Documentation," *React*, 2026.

[9] Pallets Projects, "Flask Documentation," *Flask*, 2026.

[10] MongoDB, Inc., "MongoDB Documentation," *MongoDB Documentation*, 2026.

APPENDICES

Appendix A – Detailed Calculations

A.1 Haversine Distance Calculation

The Haversine formula is used in SafeSea to calculate the distance between the vessel's current GPS position and a reference maritime boundary point.

$$d = 2Rsin^{-1}\left(\sqrt{sin^{2}\left(\frac{\Delta\phi}{2}\right) + \cos cos\, \phi_1\ \cos cos\, \phi_2\, sin^{2}\left(\frac{\Delta\lambda}{2}\right)}\right)$$

Where:

- d= distance between two locations
- R= radius of Earth
- ϕ_1, ϕ_2= latitudes
- λ_1, λ_2= longitudes
- $\Delta\phi$= difference in latitude
- $\Delta\lambda$= difference in longitude

The calculated distance is compared with the predefined safety threshold. If the vessel approaches a restricted or hazardous area, the system generates an appropriate warning.

Appendix B – Engineering Drawings / Schematics

The following OOAD and system design diagrams are included:

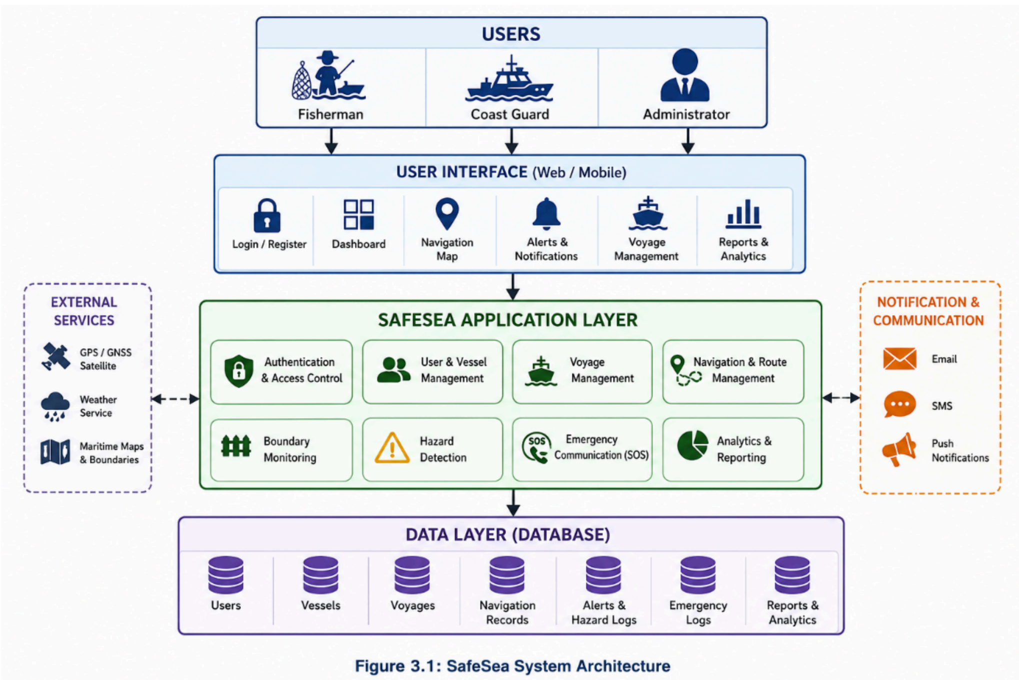


Figure B.1: SafeSea System Architecture

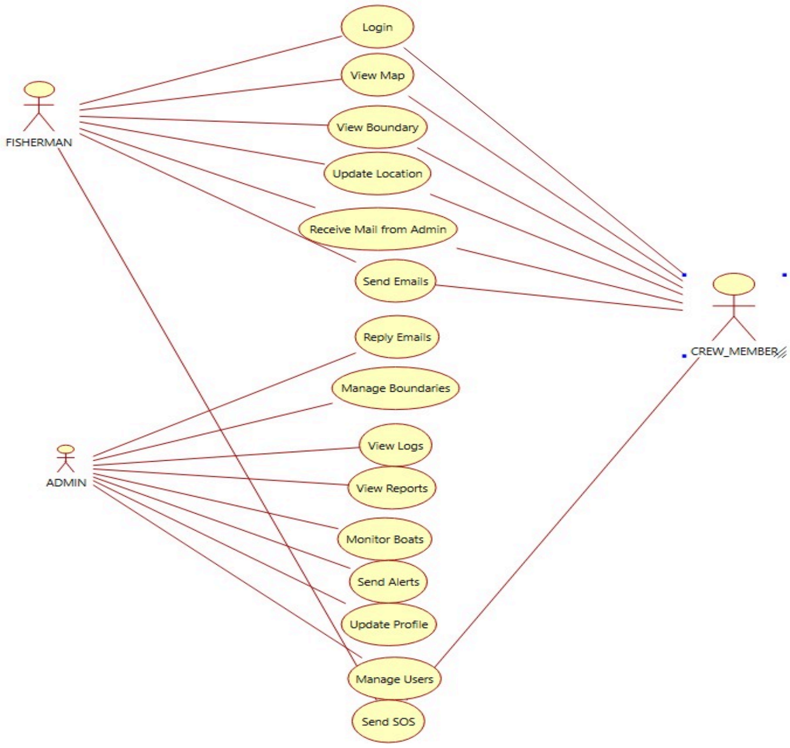


Figure B.2: SafeSea Use Case Diagram

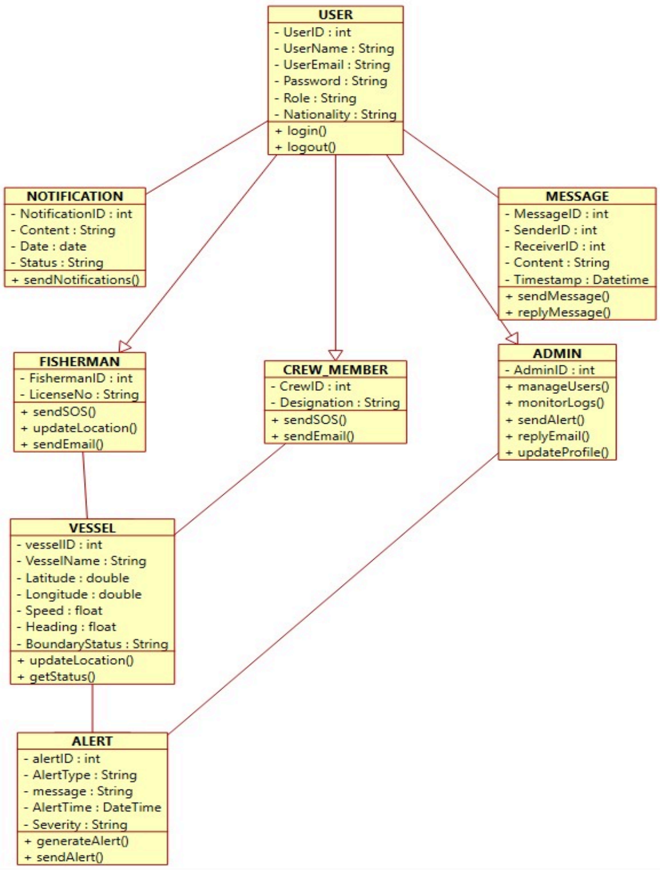


Figure B.3: SafeSea Class Diagram

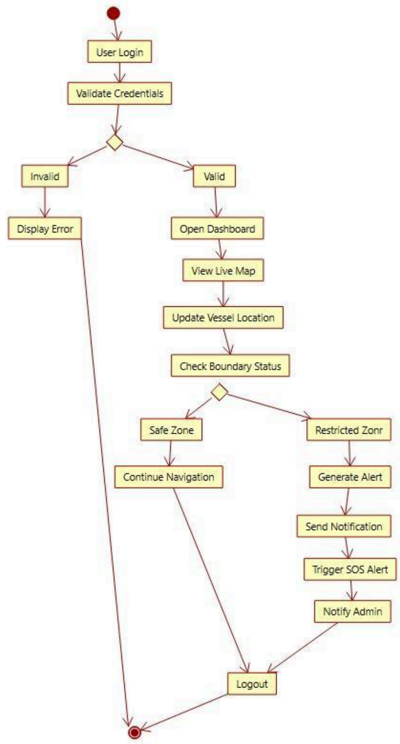
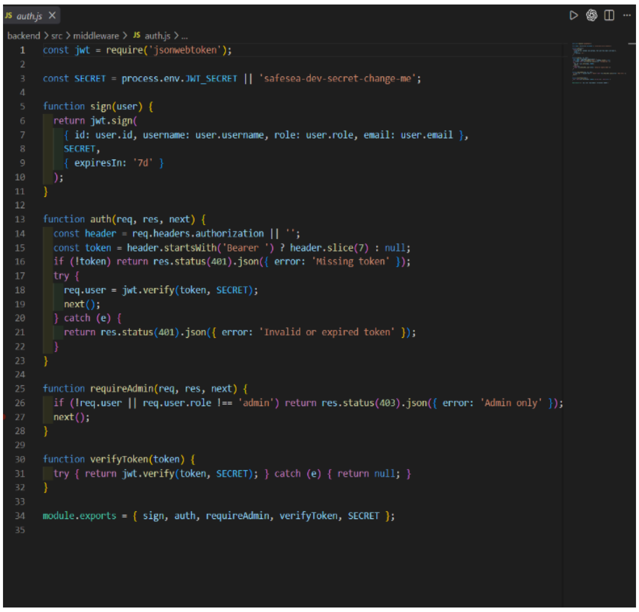


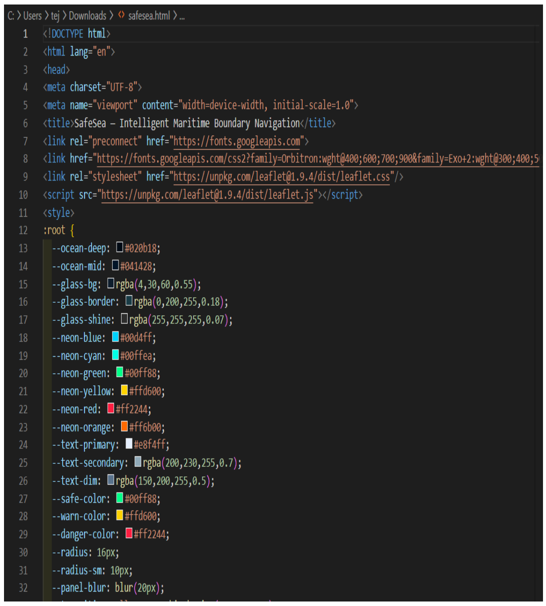
Figure B.4: SafeSea Activity Diagram

Appendix C – Source Code / Code Repository Information

The SafeSea system source code is maintained separately in the approved project repository.




```
const jwt = require('jsonwebtoken');

const SECRET = process.env.JWT_SECRET || 'safesea-dev-secret-change-me';

function sign(user) {
  return jwt.sign(
    { id: user.id, username: user.username, role: user.role, email: user.email },
    SECRET,
    { expiresIn: '7d' }
  );
}

function auth(req, res, next) {
  const header = req.headers.authorization || '';
  const token = header.startsWith('Bearer ') ? header.slice(7) : null;
  if (!token) return res.status(401).json({ error: 'Missing token' });
  try {
    req.user = jwt.verify(token, SECRET);
    next();
  } catch (e) {
    return res.status(401).json({ error: 'Invalid or expired token' });
  }
}

function requireAdmin(req, res, next) {
  if (!req.user || req.user.role !== 'admin') return res.status(403).json({ error: 'Admin only' });
  next();
}

function verifyToken(token) {
  try { return jwt.verify(token, SECRET); } catch (e) { return null; }
}

module.exports = { sign, auth, requireAdmin, verifyToken, SECRET };
```




```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SafeSea – Intelligent Maritime Boundary Navigation</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?family=Orbitron:wght@400;600;700;900&family=Exo+2:wght@300;400;5
<link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"/>
<script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
<style>
:root {
  --ocean-deep: #020b18;
  --ocean-mid: #041428;
  --glass-bg: rgba(4,30,60,0.55);
  --glass-border: rgba(0,200,255,0.18);
  --glass-shine: rgba(255,255,255,0.07);
  --neon-blue: #00d4ff;
  --neon-cyan: #00ffea;
  --neon-green: #00ff88;
  --neon-yellow: #ffd600;
  --neon-red: #ff2244;
  --neon-orange: #ff6b00;
  --text-primary: #e8f4ff;
  --text-secondary: rgba(200,230,255,0.7);
  --text-dim: rgba(150,200,255,0.5);
  --safe-color: #00ff88;
  --warn-color: #ffd600;
  --danger-color: #ff2244;
  --radius: 16px;
  --radius-sm: 10px;
  --panel-blur: blur(20px);
```

index.js

backend > src > index.js > ...

```javascript
const http = require('http');
const express = require('express');
const cors = require('cors');
const { Server } = require('socket.io');

const { pool, init } = require('./db');
const { verifyToken } = require('./middleware/auth');

const authRoutes = require('./routes/auth');
const { router: vesselRoutes, shape: vesselShape } = require('./routes/vessels');
const { router: notifRoutes } = require('./routes/notifications');
const { router: messageRoutes } = require('./routes/messages');
const usersRoutes = require('./routes/users');
const makeSosRouter = require('./routes/sos');

const PORT = parseInt(process.env.PORT || '4000', 10);

const app = express();
app.use(cors());
app.use(express.json({ limit: '1mb' }));

app.get('/health', (req, res) => res.json({ ok: true, ts: Date.now() }));

const server = http.createServer(app);
const io = new Server(server, { cors: { origin: '*' } });

// Mount REST routes
app.use('/api/auth', authRoutes);
app.use('/api/vessels', vesselRoutes);
app.use('/api/notifications', notifRoutes);
app.use('/api/messages', messageRoutes);
app.use('/api/users', usersRoutes);
app.use('/api/sos', makeSosRouter(io));

// ── Socket.IO: real-time vessel tracking ──────────────────
io.use((socket, next) => {
  // token optional — tracking feed is broadcast to connected clients
  const token = socket.handshake.auth && socket.handshake.auth.token;
  socket.user = token ? verifyToken(token) : null;
  next();
});
```

schema.sql

db > schema.sql

```sql
-- ════════════════════════════════════════════════
-- SafeSea — PostgreSQL schema
-- ════════════════════════════════════════════════

CREATE TABLE IF NOT EXISTS users (
  id            SERIAL PRIMARY KEY,
  username      TEXT UNIQUE NOT NULL,
  password_hash TEXT NOT NULL,
  role          TEXT NOT NULL CHECK (role IN ('fisherman','crew','admin')),
  fullname      TEXT NOT NULL,
  email         TEXT UNIQUE NOT NULL,
  phone         TEXT,
  vessel        TEXT,
  nationality   TEXT DEFAULT 'IND',
  created_at    TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE IF NOT EXISTS vessels (
  id          SERIAL PRIMARY KEY,
  vessel_id   TEXT UNIQUE NOT NULL,
  name        TEXT NOT NULL,
  lat         DOUBLE PRECISION NOT NULL,
  lon         DOUBLE PRECISION NOT NULL,
  spd         DOUBLE PRECISION NOT NULL DEFAULT 0,
  dist        DOUBLE PRECISION NOT NULL DEFAULT 0,
  risk        INTEGER NOT NULL DEFAULT 0,
  zone        TEXT NOT NULL DEFAULT 'safe' CHECK (zone IN ('safe','warning','danger')),
  updated_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE IF NOT EXISTS notifications (
  id          SERIAL PRIMARY KEY,
  type        TEXT NOT NULL,
  ico         TEXT NOT NULL,
  bg          TEXT NOT NULL,
  title       TEXT NOT NULL,
  message     TEXT NOT NULL,
  time_label  TEXT NOT NULL,
  unread      BOOLEAN NOT NULL DEFAULT true,
  is_sos      BOOLEAN NOT NULL DEFAULT false,
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
```

Technologies Used

Component	Technology
Frontend	React.js, JavaScript
Backend	Flask
Database	MongoDB
API Testing	Postman
Development	Visual Studio Code
Version Control	Git/GitHub

Appendix D – Experimental / Raw Data

The following functional testing records were used to evaluate the SafeSea system.

Test Category	Tests Performed	Passed	Result
Registration & Login	2	2	Pass
Voyage Management	2	2	Pass
Navigation Monitoring	2	2	Pass
Boundary & Hazard	2	2	Pass

Detection			
Emergency Communication	1	1	Pass
Access Control	1	1	Pass
Total	**10**	**10**	**Pass**

Appendix E – Ethics / Institutional Approval

SafeSea is a software-based academic project developed for educational purposes. The system does not involve human subjects, medical experiments, animal testing, or collection of sensitive research data for experimental purposes.

User and vessel information used within the system is intended to be handled through authentication and role-based access control.

Appendix F – Project Meeting / Progress Records

The project progress was reviewed through regular team discussions covering:

- Requirement identification
- Literature review
- OOAD modelling
- System architecture
- Module development
- Database integration
- Testing and debugging
- Final evaluation
- Report preparation

Progress Summary

Phase	Major Activity	Status
1	Problem Identification & Requirements	Completed
2	Literature Review	Completed
3	OOAD Modelling & Design	Completed

4	Interface & Backend Development	Complete d
5	Navigation, Boundary & Emergency Features	Complete d
6	Database Integration & Analytics	Complete d
7	Testing & Validation	Complete d
8	Final Evaluation & Documentation	Complete d

Appendix G – Individual Contribution Evidence

This appendix is **mandatory** because SafeSea is a team project.

Student	Contribution Area	Evidence
Tejaswini M (192411079)	Navigation and Safe Route Management	Navigation interface, route-management functions, testing and documentation
Sharu Priya M (192411116)	Fisherman Registration and Voyage Management	Registration, authentication, voyage functions and documentation
Vishalsai BJ (192320012)	Boundary Monitoring and Emergency Communication	Boundary detection, alerts, emergency communication and testing
D Chaitanya (192320028)	Navigation Analytics and System Evaluation	Analytics, reporting, evaluation, OOAD diagrams and documentation

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)